This line is a little tricky, so hang in there. First, we start with a `for` loop initializing a variable `i` with a value of 0 that will increment by 1 each time through the loop or three times until `i` is no longer less than the value 3, when the loop will exit. Each time through the loop we are calling the `digitalWrite()` function to turn on, or make a pin `HIGH`, as determined by the index of the array `rgb[]` that in this case is the same variable `i` as declared in the local scope of this for loop. In this way, the first time through the loop pin 9 is turned on, the second is pin 10, and the third is pin 11, when the loop will exit. This single line of code will turn on three LEDs connected to three different pins almost nearly simultaneously.

Arrays of Values

Let's say that we had previously recorded data from a sensor and now want to use these values to control the brightness of an LED or maybe alter the speed of a motor. To see this in action, let's connect a single LED with a 220 ohm resistor to digital pin 11, as shown in Figures 8-3 and 8-4.

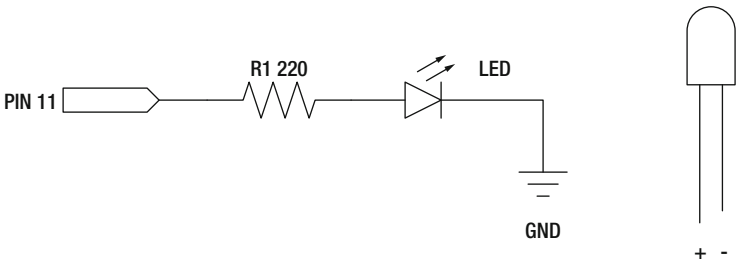


Figure 8-3. *Flicker schematic*

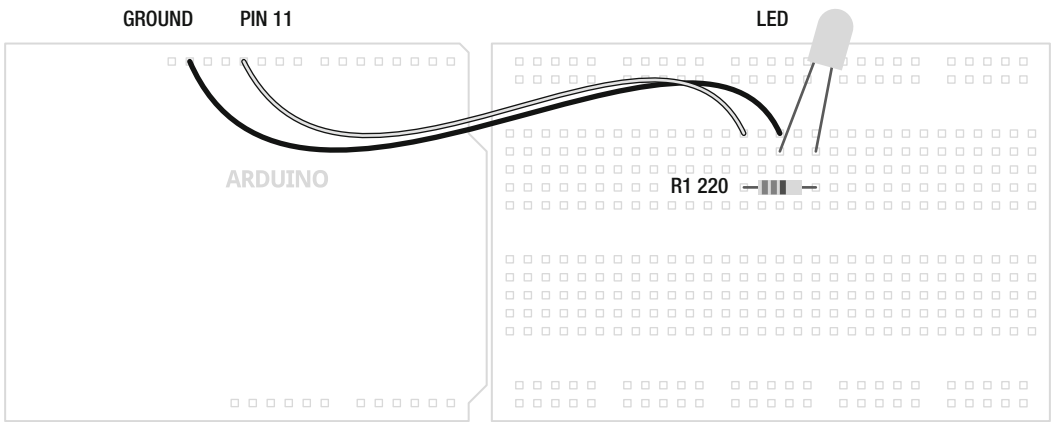


Figure 8-4. *Flicker illustration*

With this connected, Listing 8-2 provides a very simple example of how an array might work to control the brightness of a single LED connected to PWM pin 11 on the Arduino board.